

An Explainable and Clinically Interpretable Machine Learning Framework for Ethical Cardiovascular Risk Prediction

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Abstract:

Cardiovascular disease remains a major global health concern, creating a continued need for reliable and clinically understandable risk prediction systems. Although machine learning methods have demonstrated promising performance in cardiovascular prediction tasks, many existing approaches rely on complex black-box models that provide limited interpretability for clinicians and patients. In healthcare applications, predictive accuracy alone is often insufficient without transparency and responsible communication of risk estimates.

This study proposes an explainable and clinically interpretable machine learning framework for cardiovascular risk prediction using two publicly available datasets: the Framingham Heart Study dataset and the UCI Heart Disease dataset. Logistic regression was selected as the primary predictive approach because of its transparency and ability to provide clinically meaningful coefficient and odds-ratio interpretation. The framework incorporates preprocessing pipelines, missing value imputation, feature scaling, categorical encoding, and stratified five-fold cross-validation.

In addition to generating probabilistic risk estimates, the proposed system provides human-readable explanations describing contributing cardiovascular risk factors, protective indicators, and lifestyle-oriented recommendations while preserving transparent probabilistic reasoning and low computational complexity. Experimental evaluation demonstrated reliable predictive capability. The Framingham-based logistic regression model achieved a mean ROC-AUC of 0.724 (95% CI: 0.704-0.745), while the UCI Heart Disease logistic regression and random forest models achieved mean ROC-AUC values of 0.915 (95% CI: 0.885-0.945) and 0.910 (95% CI: 0.881-0.939), respectively.

Rather than emphasizing predictive performance alone, the study focuses on explainability, ethical communication, and clinical interpretability. The proposed framework highlights the potential of lightweight and transparent machine learning systems for trustworthy cardiovascular risk assessment and healthcare decision-support research.

Keywords: Cardiovascular disease prediction, explainable artificial intelligence, interpretable machine learning, logistic regression, ethical AI, healthcare informatics, cardiovascular risk assessment.

I. Introduction

Cardiovascular disease (CVD) remains one of the leading causes of mortality worldwide and continues to impose a major burden on healthcare systems. Conditions such as coronary artery disease, heart failure, and stroke contribute significantly to long-term disability and premature death across diverse populations. Early identification of individuals at elevated cardiovascular risk is therefore an important objective in preventive medicine and clinical decision support.

The increasing availability of electronic health records and publicly accessible clinical datasets has encouraged the use of machine learning techniques for cardiovascular risk prediction. Methods including logistic regression, support vector machines, decision trees, ensemble learning approaches, and deep neural networks have demonstrated promising predictive performance in identifying cardiovascular abnormalities and estimating disease risk. However, many highly accurate machine learning systems operate as black-box models, providing limited interpretability regarding how predictions are generated. In healthcare settings, lack of transparency may reduce clinician trust and restrict practical adoption of predictive systems.

Explainability and clinical interpretability have consequently become important research priorities in biomedical informatics. Clinicians often require understandable reasoning linked to recognized medical risk factors before relying on predictive outputs in clinical decision-making. Similarly, patient-centered healthcare applications benefit from explanations that clearly describe contributing and protective cardiovascular factors. Transparent models capable of relating predictions to clinically meaningful variables may therefore provide advantages in communication and usability.

Another important consideration involves ethical communication of predictive results. Healthcare AI systems should not be presented as replacements for professional diagnosis or clinical judgment. Responsible risk prediction frameworks must acknowledge uncertainty, communicate limitations clearly, and avoid overstated diagnostic claims. These concerns are particularly relevant in cardiovascular prediction, where automated outputs may directly influence patient perception and healthcare decisions.

Motivated by these considerations, this study proposes an explainable and clinically interpretable machine learning framework for cardiovascular risk prediction using the Framingham Heart Study dataset and the UCI Heart Disease dataset. Logistic regression was selected as the primary predictive approach because it provides direct interpretability through coefficient and odds-ratio analysis while maintaining reliable predictive capability. The framework incorporates preprocessing pipelines, stratified cross-validation, and human-readable explanation strategies describing contributing risk factors, protective indicators, and lifestyle-oriented guidance.

The major contributions of the present study are summarized as follows:

1. Development of an explainable and clinically interpretable cardiovascular prediction framework using two clinically distinct public cardiovascular datasets;
2. Integration of lightweight logistic regression modeling with human-readable cardiovascular risk explanation strategies designed to support transparent clinical communication;
3. Comparative evaluation between interpretable linear models and ensemble-based approaches to analyze the trade-off between predictive capability and explainability;
4. Incorporation of ethically framed risk communication, including transparent presentation of probabilistic predictions, contributing risk factors, protective indicators, and non-diagnostic medical guidance;

5. Emphasis on low-complexity and clinically accessible prediction strategies suitable for preventive healthcare and resource-constrained environments.

Unlike approaches focused primarily on maximizing classification accuracy, the proposed work emphasizes transparency, ethical communication, and clinically meaningful interpretation. The study aims to contribute toward the development of lightweight and human-centered machine learning systems for trustworthy cardiovascular risk assessment.

II. Related Work

A. Traditional Cardiovascular Risk Assessment Models

Cardiovascular risk prediction has been an important area of clinical research for several decades. One of the earliest and most influential epidemiological investigations of cardiovascular disease was the Framingham Heart Study (Dawber et al., 1951). Building on this landmark study, the Framingham Risk Score, developed by Wilson et al. (1998), estimates long-term cardiovascular risk using factors such as age, blood pressure, cholesterol levels, smoking status, and diabetes. Other commonly used systems, including SCORE, QRISK, and ACC/AHA risk calculators, have also supported preventive cardiology and clinical screening practices.

Traditional statistical models remain valuable because they are relatively transparent and clinically interpretable. Clinicians can generally understand how these systems generate predictions and relate outputs to established cardiovascular risk factors. However, conventional scoring systems may show limited adaptability when applied to evolving clinical datasets or different patient populations. These limitations have encouraged exploration of machine learning approaches capable of learning more flexible relationships from healthcare data.

B. Machine Learning Approaches for Cardiovascular Disease Prediction

The growing availability of electronic health records and public clinical datasets has increased the use of machine learning techniques for cardiovascular disease prediction. Supervised learning algorithms including logistic regression, decision trees, support vector machines, random forests, gradient boosting methods, and neural networks have been widely investigated for heart disease classification and cardiovascular risk estimation.

Logistic regression continues to be commonly used because of its simplicity, computational efficiency, and direct interpretability. More advanced approaches such as ensemble learning and deep learning models have demonstrated improved predictive capability in some studies, particularly when complex nonlinear relationships exist among variables. Deep learning methods have also been explored for large-scale healthcare analytics involving imaging data and physiological signals.

Despite promising predictive performance, highly complex machine learning systems often provide limited interpretability regarding how predictions are generated. In healthcare settings, lack of understandable reasoning may reduce practical clinical usability. As a result, recent cardiovascular prediction research has increasingly emphasized the importance of interpretable and transparent machine learning frameworks.

C. Explainable Artificial Intelligence in Healthcare

Explainable Artificial Intelligence (XAI) has emerged as an important area of healthcare machine learning research. Since predictive systems may influence medical decisions and patient management, clinicians often prefer models that provide understandable reasoning rather than unexplained outputs. Recent studies have emphasized that successful integration of artificial intelligence into healthcare requires balancing predictive performance with transparency, clinical usability, and human-centered decision support (Topol, 2019). The growing importance of explainable AI in healthcare has also been emphasized in recent biomedical research (Esteva et al., 2019). Furthermore, interpretability has become a major research objective in trustworthy artificial intelligence, with increasing emphasis on developing models that are both accurate and understandable (Doshi-Velez & Kim, 2017). The importance of interpretable machine learning for healthcare has also been highlighted by Ahmad et al. (2018), who emphasized that transparent predictive models can improve clinicians' confidence and support informed medical decision-making.

Several explanation techniques have been proposed to improve transparency of machine learning systems. Approaches such as LIME (Ribeiro et al., 2016) and SHAP (Lundberg & Lee, 2017) are commonly used to estimate feature contributions and analyse the influence of variables on prediction outcomes. At the same time, inherently interpretable models such as logistic regression remain important because they allow direct interpretation without requiring additional explanation frameworks. Logistic regression coefficients and odds ratios can be naturally associated with recognized cardiovascular risk factors, improving clinical readability of model outputs.

In cardiovascular prediction tasks, explainability is particularly useful because many contributing factors, including smoking, hypertension, obesity, diabetes, and exercise-related symptoms, are already medically well understood. Transparent prediction models may therefore improve communication of cardiovascular risk estimates for both clinicians and patients.

D. Ethical Considerations in Machine Learning-Based Healthcare Systems

As machine learning systems become increasingly integrated into healthcare research, machine learning has demonstrated considerable potential in medicine, while also highlighting the need for transparency and responsible deployment (Rajkomar et al., 2019). Ethical concerns related to transparency, accountability, and responsible communication have therefore gained significant attention. Predictive models developed using retrospective datasets may contain demographic limitations, sampling bias, or incomplete clinical information, which can affect generalizability and reliability across broader healthcare settings.

Another important concern involves communication of AI-generated predictions. Black-box systems may provide highly accurate outputs while offering limited explanation regarding why a patient is categorized as high risk. In healthcare applications, lack of transparency may reduce trust and complicate interpretation of predictive results. Consequently, recent studies have emphasized the importance of interpretable models, transparent reasoning, and responsible communication of uncertainty in healthcare AI systems.

Ethically responsible cardiovascular prediction frameworks should therefore balance predictive capability with explainability, clinical usability, and appropriate medical framing of risk estimates.

E. Research Gap and Motivation for the Present Study

Although considerable progress has been made in cardiovascular disease prediction using machine learning, several important limitations remain in existing research. Many studies focus primarily on maximizing predictive accuracy through increasingly complex computational models while giving comparatively less attention to interpretability and clinically understandable communication of results.

In addition, several cardiovascular prediction systems present outputs mainly as probability estimates without sufficient explanation of contributing risk factors, protective indicators, or practical lifestyle-oriented guidance. Ethical framing and transparent communication of uncertainty also remain underemphasized in many healthcare AI studies.

The present study addresses these gaps by proposing a lightweight and clinically interpretable cardiovascular prediction framework using two publicly available cardiovascular datasets. Unlike approaches centered mainly on predictive complexity, the proposed framework emphasizes transparent probabilistic reasoning, human-readable explanation strategies, and ethically responsible communication of cardiovascular risk estimates.

III. Materials and Datasets

A. Overview of the Study Datasets

This study utilized two publicly available cardiovascular datasets to develop and evaluate explainable machine learning models for cardiovascular risk assessment. The selected datasets represent two related but clinically distinct prediction settings. The first dataset, derived from the Framingham Heart Study, was used for long-term cardiovascular risk estimation, while the second dataset, the UCI Heart Disease dataset, was used for heart disease probability assessment.

The use of two separate datasets allowed the study to evaluate model behavior under different clinical prediction scenarios while maintaining a consistent emphasis on interpretability and ethical risk communication. Both datasets are widely used in cardiovascular machine learning research and provide structured clinical variables relevant to cardiovascular risk analysis. The major characteristics of the datasets used in the present study are summarized in Table I.

Table I

Characteristics of the Cardiovascular Datasets Used in the Proposed Explainable Machine Learning Framework

Dataset	Clinical Purpose	Number of Features Used	Target Variable	Prediction Type	Machine Learning Models	Key Clinical Variables
Framingham Heart Study Dataset	Long-term cardiovascular risk estimation	6	Ten Year CHD	10-year cardiovascular risk prediction	Logistic Regression	Age, sex, systolic blood pressure, smoking status, diabetes, BMI

UCI Heart Disease Dataset	Heart disease probability assessment	13	Heart disease presence/absence	Cross-sectional heart disease classification	Logistic Regression, Random Forest	Chest pain type, cholesterol, ECG findings, exercise-induced angina, ST depression, blood pressure
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B. Framingham Heart Study Dataset

The Framingham Heart Study dataset was used to develop the basic cardiovascular risk assessment model. The dataset originates from the well-known Framingham Heart Study, a long-term epidemiological research initiative focused on identifying factors associated with cardiovascular disease development. The dataset contains demographic, lifestyle, and clinical information collected from participants over multiple years.

For the present study, the prediction objective involved estimating the probability of developing coronary heart disease within a ten-year period. To preserve model simplicity and clinical interpretability, only a limited subset of clinically accessible variables was selected. These variables included sex, age, systolic blood pressure, smoking status, diabetes status, and body mass index. The target variable represented the ten-year coronary heart disease outcome.

The selected feature set was intentionally restricted to variables that are commonly available during routine clinical assessment without requiring advanced laboratory testing or specialized imaging procedures. This lightweight feature selection strategy was adopted to support transparency, accessibility, and practical interpretability of the resulting prediction model.

Because the dataset contained missing values in several variables, missing data handling was incorporated directly into the preprocessing pipeline through median imputation. The Framingham-based model was evaluated using stratified five-fold cross-validation and receiver operating characteristic area under the curve (ROC-AUC) analysis.

C. UCI Heart Disease Dataset

The second dataset used in this study was the UCI Heart Disease dataset, obtained from the UCI Machine Learning Repository (Dua & Graff, 2019), which is widely used for machine learning research involving heart disease classification. Unlike the Framingham dataset, which focuses on longitudinal cardiovascular risk estimation, the UCI dataset is primarily designed for predicting the presence or absence of heart disease based on clinical measurements and diagnostic indicators.

The dataset contains demographic, physiological, electrocardiographic, and exercise-related attributes associated with cardiovascular health. Variables included age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting electrocardiographic findings, maximum heart rate achieved, exercise-induced angina, ST depression, slope of the ST segment, number of major vessels, and thalassemia-related indicators. The target variable was converted into a binary classification task representing heart disease presence or absence.

Since the dataset included categorical variables and missing entries represented by placeholder symbols, additional preprocessing procedures were required. Missing values were replaced and handled using imputation strategies, while categorical variables were transformed using one-hot encoding. Numerical variables were standardized prior to model training.

The UCI dataset enabled evaluation of the proposed explainable framework under a more clinically detailed prediction setting involving a broader range of cardiovascular indicators. In addition to logistic regression, a random forest classifier was also evaluated as a nonlinear benchmark model for comparative analysis. The clinical variables selected for cardiovascular prediction across both datasets are presented in Table II.

Table II
Clinical Features Used for Cardiovascular Risk Prediction

Feature	Dataset	Clinical Description	Variable Type
Sex	Framingham, UCI	Biological sex of the patient	Categorical
Age	Framingham, UCI	Age in years	Numerical
Systolic Blood Pressure (sysBP / trestbps)	Framingham, UCI	Resting systolic blood pressure measurement	Numerical
Smoking Status	Framingham	Current smoking behavior	Categorical
Diabetes Status	Framingham	Presence or absence of diabetes mellitus	Categorical
Body Mass Index (BMI)	Framingham	Body weight relative to height	Numerical
Chest Pain Type (cp)	UCI	Type of chest pain experienced by the patient	Categorical
Serum Cholesterol (chol)	UCI	Cholesterol concentration in blood	Numerical
Fasting Blood Sugar (fbs)	UCI	Fasting blood sugar level indicator	Categorical
Resting ECG Results (restecg)	UCI	Electrocardiographic findings at rest	Categorical
Maximum Heart Rate (thalach)	UCI	Maximum heart rate achieved during exercise	Numerical
Exercise-Induced Angina (exang)	UCI	Presence of angina during exercise	Categorical
ST Depression (oldpeak)	UCI	ST-segment depression induced by exercise	Numerical
ST Segment Slope (slope)	UCI	Slope of peak exercise ST segment	Categorical
Number of Major Vessels (ca)	UCI	Number of major vessels identified through fluoroscopy	Numerical
Thalassemia Indicator (thal)	UCI	Thalassemia-related clinical finding	Categorical

D. Data Preprocessing and Quality Handling

Appropriate preprocessing procedures were applied to both datasets to ensure consistency and improve model reliability. Data preprocessing was implemented using end-to-end machine learning pipelines to reduce manual intervention and maintain reproducibility.

For numerical variables, missing values were handled using median imputation followed by feature standardization. Median imputation was selected because it provides robustness against outliers and preserves central tendencies within clinical measurements. Standardization was applied to normalize feature scales prior to model training.

For categorical variables in the UCI dataset, missing entries were handled using the most frequent value imputation strategy. One-hot encoding was then applied to transform categorical variables into machine-readable numerical representations while preserving category distinctions.

The preprocessing framework was implemented using scikit-learn machine learning libraries (Pedregosa et al., 2011) and column transformation strategies to ensure systematic handling of mixed data types and reduce the likelihood of preprocessing inconsistencies between training and evaluation stages.

E. Ethical and Data Usage Considerations

Both datasets used in this study are publicly available and commonly utilized for educational and research purposes within the machine learning and biomedical informatics communities. The datasets do not contain personally identifiable information, and no patient re-identification procedures were involved in this work.

Although publicly available datasets provide valuable opportunities for methodological research, they also contain important limitations. The datasets represent specific populations, collection periods, and clinical environments, which may limit generalizability to broader or more diverse populations. Consequently, the predictive models developed in this study should not be interpreted as universally applicable clinical diagnostic systems.

The present work is intended primarily as a research-oriented and educational exploration of explainable cardiovascular risk prediction rather than a replacement for professional medical evaluation or diagnosis. Ethical communication and transparent acknowledgment of model limitations were therefore considered important aspects of the study design.

IV. Methodology

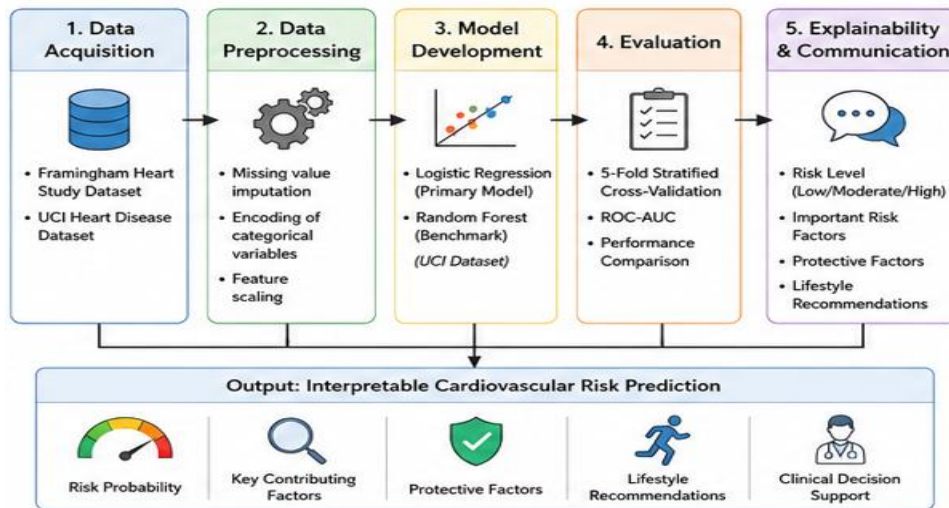
A. Overall Study Framework

The proposed framework was designed to develop explainable and clinically interpretable machine learning models for cardiovascular risk prediction using publicly available clinical datasets. The overall methodology consisted of data acquisition, preprocessing, feature selection, model development, performance evaluation, and explainability-oriented risk communication.

Two separate prediction settings were investigated in this study. The first involved long-term cardiovascular risk estimation using the Framingham Heart Study dataset, while the second focused on heart disease probability assessment using the UCI Heart Disease dataset. Although both tasks are related to cardiovascular prediction, they represent clinically distinct objectives and therefore required separate modeling approaches.

The complete workflow of the proposed system consisted of the following stages: data preprocessing, handling of missing values, feature transformation, machine learning model training, cross-validation-based evaluation, probability prediction, and generation of human-readable explanations describing contributing and protective clinical factors. The framework was implemented using reproducible end-to-end machine learning pipelines. The overall workflow of the proposed explainable cardiovascular prediction framework is illustrated in Fig. 1.

Fig. 1. Overall methodology and workflow of the study



B. Data Preprocessing

Appropriate preprocessing procedures were applied to both datasets before model development. Since clinical datasets frequently contain incomplete or heterogeneous information, preprocessing was considered an essential step for improving model stability and reliability.

For the Framingham dataset, the selected clinical variables included demographic and physiological measurements such as sex, age, systolic blood pressure, smoking status, diabetes status, and body mass index. Missing numerical values were handled using median imputation to reduce the effect of incomplete records while preserving the distribution of clinical measurements. After imputation, numerical variables were standardized using feature scaling techniques prior to model training.

The preprocessing strategy for the UCI Heart Disease dataset was comparatively more extensive because the dataset contained both numerical and categorical variables, along with missing entries represented by placeholder symbols. Missing values were first converted into numerical null representations before preprocessing. Numerical variables were processed using median imputation followed by standardization, while categorical variables were handled using most-frequent-value imputation and one-hot encoding. These transformations were implemented using a column transformation pipeline to ensure systematic preprocessing of mixed data types.

The use of preprocessing pipelines reduced the possibility of inconsistencies between training and evaluation stages and improved reproducibility of the overall methodology.

C. Feature Selection Strategy

Feature selection in the present study was guided not only by predictive capability but also by clinical accessibility and interpretability. Rather than incorporating a very large number of variables, the study intentionally prioritized clinically meaningful and easily obtainable features.

For the Framingham-based model, only a limited set of basic cardiovascular risk indicators was selected. These included age, sex, systolic blood pressure, smoking status, diabetes status, and body mass index. The purpose of restricting the feature set was to preserve transparency while allowing the model to operate using information commonly available during routine clinical assessment. This lightweight design approach also improves the practical accessibility of the framework in low-resource or preventive healthcare settings.

The UCI Heart Disease dataset allowed the inclusion of additional cardiovascular indicators such as chest pain type, electrocardiographic findings, exercise-induced angina, maximum heart rate achieved, and ST-segment depression measurements. These variables provided a more clinically detailed representation of cardiovascular status while still maintaining interpretability within the selected modeling framework.

The feature selection strategy therefore balanced predictive relevance with the broader objective of explainable and clinically interpretable machine learning.

D. Machine Learning Models

1) Logistic Regression Model

Logistic regression was selected as the primary predictive model for both datasets. It remains one of the most widely accepted statistical methods for binary outcome prediction in medical research (Hosmer et al., 2013). Although more complex machine learning approaches are available, logistic regression offers several advantages in healthcare applications, particularly in terms of transparency and interpretability.

The model estimates the probability of cardiovascular risk or heart disease presence by learning relationships between clinical variables and binary target outcomes. Because logistic regression coefficients can be directly interpreted, the contribution of individual clinical variables to predicted risk can be analyzed in a meaningful manner. Odds ratios derived from model coefficients further support clinical interpretation of feature importance.

The logistic regression implementation utilized iterative optimization with standardized input variables and was integrated into end-to-end preprocessing pipelines to ensure reproducible training procedures.

2) Random Forest Benchmark Model

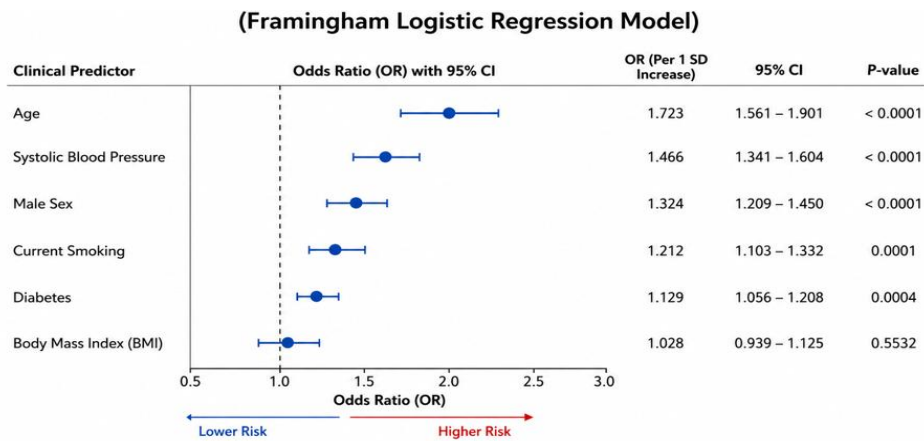
In addition to logistic regression, a random forest classifier was implemented for the UCI Heart Disease dataset as a nonlinear benchmark model. Random Forest was introduced by Breiman (2001) and is one of the most widely used ensemble learning algorithms. Random forest algorithms combine multiple decision trees to improve predictive capability and capture nonlinear interactions between variables.

The purpose of including the random forest model was not to replace the interpretable logistic regression framework, but rather to provide comparative insight regarding performance differences between transparent linear models and more complex ensemble-based approaches. The random forest classifier was trained using multiple decision trees with parallel computation support and fixed random initialization for reproducibility.

While ensemble methods may improve predictive accuracy in certain settings, they generally provide lower direct interpretability compared to logistic regression models. This trade-off

between performance and explainability was considered an important aspect of the study. The relative influence of the selected Framingham clinical variables is illustrated in Fig. 2 using odds-ratio-based interpretation.

Fig. 2. Feature importance (odds ratio) from the Framingham logistic regression model



E. Cross-Validation and Performance Evaluation

To ensure reliable evaluation of model performance, stratified five-fold cross-validation was applied during training and testing procedures. Stratified sampling was selected to preserve class distribution across folds and reduce evaluation bias associated with imbalanced clinical datasets.

The primary evaluation metric used in this study was the Receiver Operating Characteristic Area Under the Curve (ROC-AUC). ROC-AUC was selected because it measures the ability of the model to discriminate between positive and negative outcome classes across different classification thresholds. This metric is widely used in biomedical prediction research and is particularly suitable for healthcare classification problems involving probabilistic outputs.

For the Framingham-based model, cross-validation evaluation demonstrated a mean ROC-AUC of 0.724 (95% CI: 0.704-0.745). The UCI logistic regression model achieved a mean ROC-AUC of 0.915 (95% CI: 0.885-0.945), while the UCI random forest model achieved a mean ROC-AUC of 0.910 (95% CI: 0.881-0.939). Confidence intervals were estimated using the t-distribution under stratified five-fold cross-validation to quantify uncertainty and model stability.

Standard deviation values across folds were also analyzed to evaluate stability and consistency of model performance. In addition to mean ROC-AUC evaluation, variability across cross-validation folds was analyzed to assess model stability and robustness. Lower standard deviation values indicated consistent predictive behavior across different data partitions, supporting the reliability of the proposed framework under repeated stratified sampling conditions.

F. Explainability-Oriented Risk Communication

A major focus of the proposed framework involved the generation of interpretable and human-readable explanations alongside prediction outputs. Instead of presenting only numerical probability estimates, the system was designed to communicate clinically relevant contributing factors, protective indicators, and practical lifestyle-oriented guidance.

For both datasets, model outputs were categorized into clinically understandable risk levels such as low, moderate, and high risk. The framework then identified variables associated with elevated cardiovascular risk, including hypertension, smoking status, obesity, diabetes, abnormal cholesterol levels, exercise-induced angina, and age-related risk increases. Protective factors such as healthy blood pressure ranges, absence of smoking, and favorable exercise capacity were also communicated to the user.

The explainability strategy was intentionally rule-guided and clinically understandable rather than dependent on highly technical explanation mechanisms. This approach was selected to improve accessibility for both clinicians and non-technical users while preserving transparency of the decision-support process.

G. Ethical Clinical Framing

Ethical considerations were incorporated throughout the proposed framework, particularly in relation to communication of predictive outputs. The system was designed as a supportive research-oriented decision-assistance framework rather than an autonomous diagnostic tool.

Clear medical disclaimers were included to indicate that model predictions should not replace professional clinical evaluation or medical judgment. The framework avoided definitive diagnostic claims and instead presented outputs as probabilistic estimates derived from machine learning analysis of clinical variables.

This emphasis on responsible communication was considered important because healthcare AI systems may influence patient perception and clinical interpretation. The proposed methodology therefore prioritizes transparency, interpretability, and ethical framing alongside predictive performance.

V. Experimental Results

A. Evaluation Overview

The experimental analysis was conducted to evaluate the predictive capability, stability, and interpretability of the proposed cardiovascular risk prediction framework. Separate experiments were performed for the Framingham Heart Study dataset and the UCI Heart Disease dataset because the two datasets represent different clinical prediction settings.

The Framingham-based model focused on long-term cardiovascular risk estimation using a limited set of clinically accessible variables, whereas the UCI-based model focused on heart disease probability assessment using a broader set of cardiovascular indicators. Model evaluation emphasized not only predictive performance but also interpretability and transparency of generated outputs.

Stratified five-fold cross-validation was applied during model evaluation to preserve class distribution and reduce the likelihood of biased performance estimates. Receiver Operating Characteristic Area Under the Curve (ROC-AUC) was selected as the primary evaluation metric because of its suitability for probabilistic healthcare classification tasks. Comparative performance evaluation results for the developed machine learning models are summarized in Table III.

Table III**Performance Comparison of the Proposed Machine Learning Models**

Model	Dataset	Mean ROC-AUC	95% CI	Brier Score	Interpretability
Logistic Regression	Framingham	0.724	0.704–0.745	0.117	High
Logistic Regression	UCI	0.915	0.885–0.945	—	High
Random Forest	UCI	0.910	0.881–0.939	—	Moderate

B. Framingham-Based Cardiovascular Risk Model

The Framingham-based model was developed using logistic regression with a lightweight and clinically interpretable feature set consisting of age, sex, systolic blood pressure, smoking status, diabetes status, and body mass index. Missing numerical values were handled using median imputation, followed by feature standardization within the preprocessing pipeline.

Cross-validation analysis demonstrated a mean ROC-AUC of 0.724 (95% CI: 0.704-0.745), indicating moderate discriminative capability for long-term cardiovascular risk estimation. The obtained performance suggests that even a relatively small set of clinically accessible variables can provide meaningful predictive information regarding future cardiovascular risk.

Although the predictive performance of the Framingham-based model was lower than that of the UCI-based model, the simpler feature set provided important advantages in terms of interpretability and accessibility. Because the selected variables are commonly available during routine clinical assessment, the resulting framework may be more practical for transparent preventive risk communication and educational applications.

The experimental findings also highlighted the value of clinically interpretable modeling strategies in situations where model transparency may be preferred over maximum predictive complexity.

C. Calibration Analysis of the Framingham-Based Model

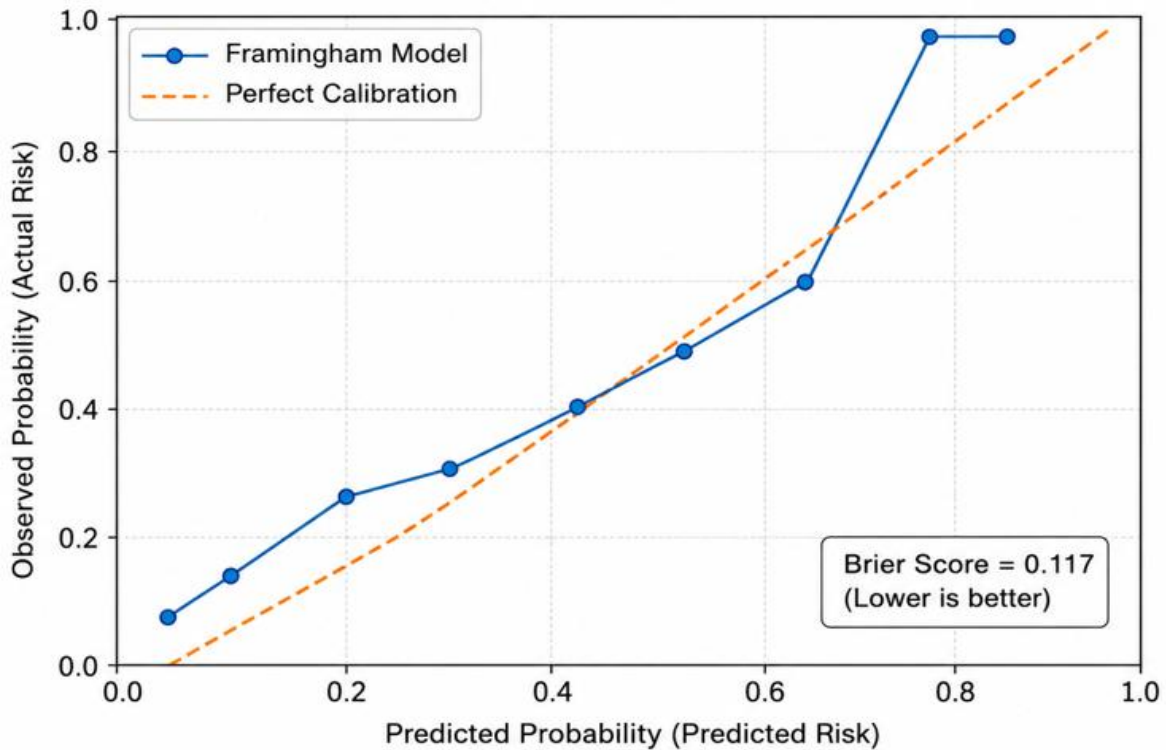
To evaluate the reliability of predicted cardiovascular risk probabilities, calibration analysis was performed for the Framingham-based logistic regression model. In addition to discriminative performance evaluation using ROC-AUC, probabilistic calibration was assessed using the Brier score and calibration curve analysis.

The Framingham model achieved a Brier score of 0.117, indicating good agreement between predicted probabilities and observed cardiovascular outcomes. The calibration curve demonstrated that predicted probabilities closely followed the ideal calibration line across most probability ranges, suggesting that the model provides reasonably reliable cardiovascular risk estimates.

Minor deviations were observed at higher predicted probability ranges, which may be attributable to the smaller number of high-risk observations. Overall, the calibration analysis supports the suitability of the proposed interpretable framework for transparent

cardiovascular risk estimation and clinical decision-support applications. The corresponding calibration curve is presented in Fig. 3.

Fig. 3. Calibration curve of the Framingham-based logistic regression model. The predicted cardiovascular risk probabilities demonstrate good agreement with observed outcomes across most probability ranges. The model achieved a Brier score of 0.117, indicating satisfactory probabilistic calibration.



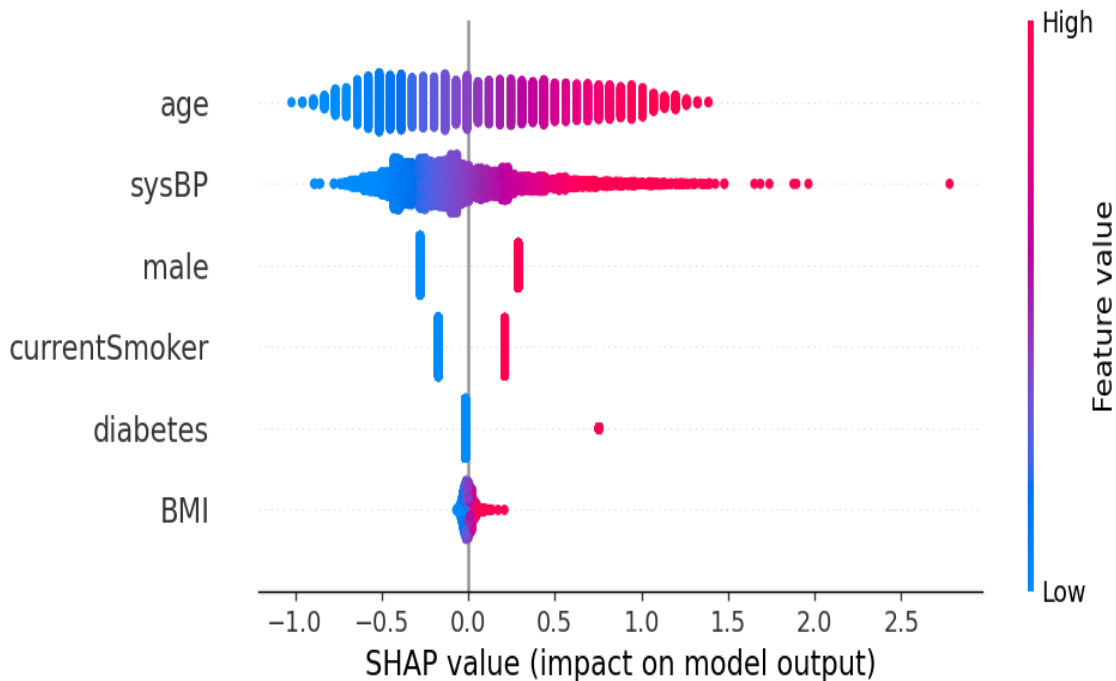
D. SHAP-Based Explainability Analysis

To complement the coefficient-based interpretation provided by logistic regression, SHAP (SHapley Additive exPlanations), introduced by Lundberg and Lee (2017), was used to quantify the contribution of each predictor variable to cardiovascular risk prediction. The SHAP summary plot demonstrated that age and systolic blood pressure were the most influential predictors in the Framingham-based model, followed by sex, smoking status, and diabetes.

Higher values of age and systolic blood pressure were associated with increased predicted cardiovascular risk, consistent with established clinical evidence. Smoking and diabetes also contributed positively to risk prediction, whereas BMI exhibited comparatively lower influence in the present model. The agreement between SHAP explanations and the estimated odds ratios further supports the interpretability and clinical validity of the proposed framework.

The SHAP analysis provides an additional layer of transparent model interpretation, thereby enhancing clinician trust and supporting explainable decision-making in cardiovascular risk assessment. The SHAP summary plot illustrating the contribution of individual predictor variables is shown in Fig. 4.

Fig. 4. SHAP summary plot for the Framingham-based logistic regression model showing the contribution of individual predictor variables to cardiovascular risk estimation. Age and systolic blood pressure were identified as the most influential predictors.



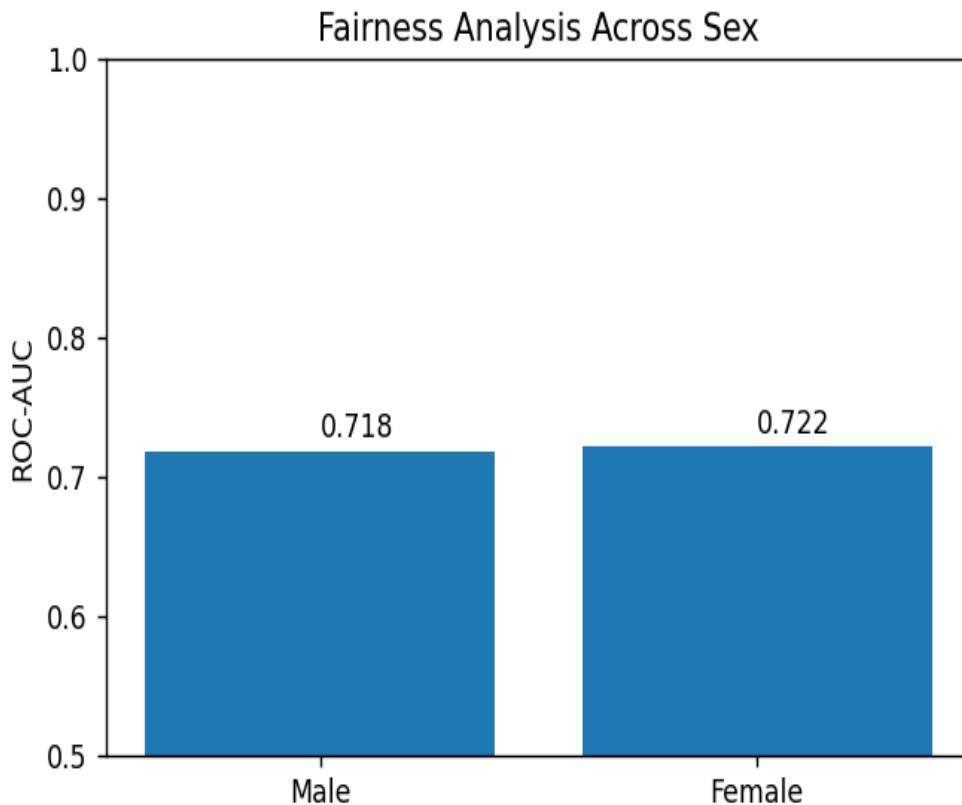
E. Fairness Analysis Across Sex

To investigate potential demographic bias, fairness analysis was conducted by evaluating predictive performance separately for male and female participants. The Framingham-based logistic regression model achieved a ROC-AUC of 0.718 for male participants and 0.722 for female participants, resulting in a performance difference of only 0.004.

The negligible difference in predictive performance suggests that the proposed model demonstrates a high degree of fairness across sex categories. These findings support the ethical suitability of the proposed interpretable machine learning framework for cardiovascular risk assessment and indicate that the model does not exhibit substantial performance disparities between male and female populations.

The fairness analysis complements the explainability and calibration evaluations performed in this study and contributes to the development of trustworthy and responsible artificial intelligence systems for healthcare applications. The fairness analysis across male and female participants is illustrated in Fig. 5.

Fig. 5. Fairness analysis of the Framingham-based logistic regression model across sex categories. The model achieved comparable predictive performance for male (ROC-AUC = 0.718) and female (ROC-AUC = 0.722) participants, indicating minimal demographic performance disparity.



F. External Validation and Generalizability Assessment

To evaluate the generalizability of the proposed interpretable cardiovascular risk prediction framework, external validation was performed using the publicly available Kaggle Cardiovascular Disease dataset comprising approximately 70,000 participants. Variables available in the external dataset were mapped to the six predictor variables used in the Framingham-based model, namely sex, age, systolic blood pressure, smoking status, diabetes status, and body mass index.

Five-fold cross-validation on the external dataset yielded a mean ROC-AUC of 0.776 with a 95% confidence interval of 0.773–0.780, indicating robust predictive performance across folds. The low standard deviation (0.004) further demonstrates the stability of the proposed model.

Calibration analysis of the external validation dataset produced a Brier score of 0.196. The calibration curve demonstrated reasonable agreement between predicted and observed cardiovascular risk probabilities across most probability ranges, supporting the reliability of the proposed framework when applied to an independent population.

The external validation results suggest that the proposed interpretable machine learning framework exhibits promising generalizability across cardiovascular populations. The external validation calibration curve is presented in Fig. 6.

Fig. 6. Calibration curve obtained during external validation using the Kaggle Cardiovascular Disease dataset. The model demonstrated reasonable agreement between predicted and observed cardiovascular risk probabilities, with a Brier score of 0.196.

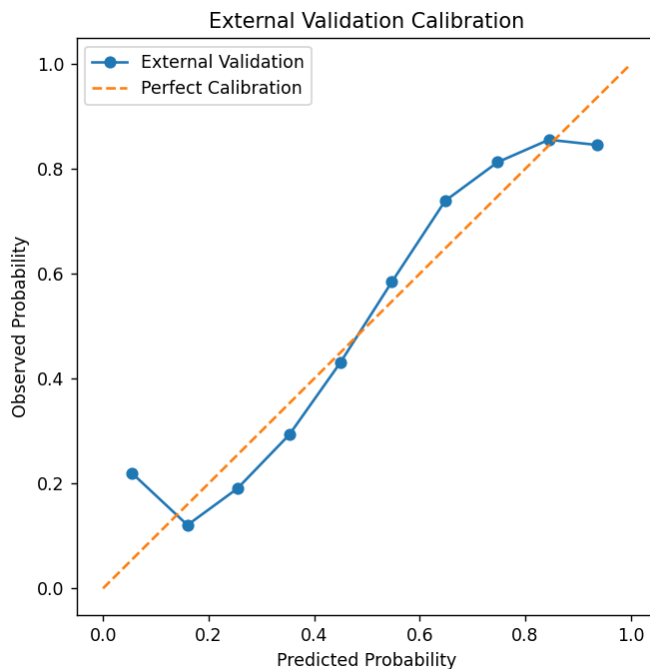


Table IV

External Validation Results

Dataset	Sample Size	Mean ROC-AUC	95% CI	Std. Dev.	Brier Score
Kaggle Cardiovascular Disease	70,000	0.776	0.773–0.780	0.004	0.196

G. UCI Heart Disease Prediction Model

The UCI Heart Disease dataset enabled evaluation of the proposed framework using a broader collection of cardiovascular indicators, including chest pain characteristics, electrocardiographic findings, exercise-induced symptoms, and vascular measurements. Both logistic regression and random forest classifiers were evaluated for comparative analysis.

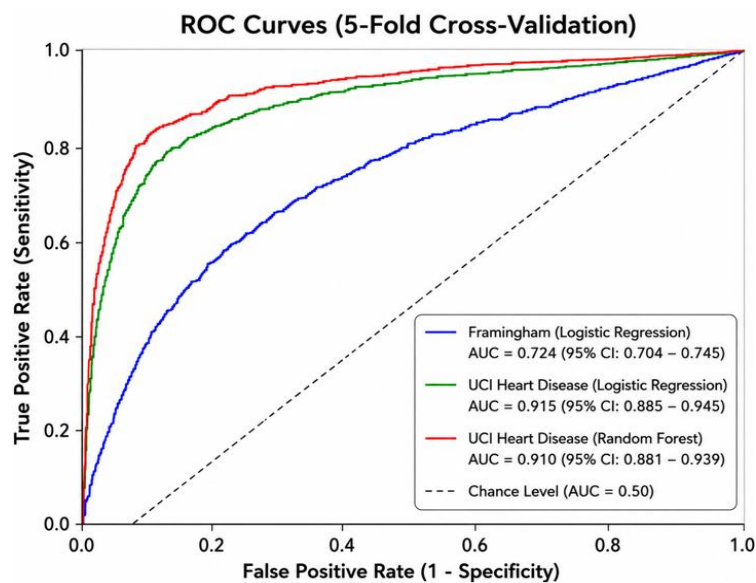
The logistic regression model achieved a mean ROC-AUC of 0.915 (95% CI: 0.885-0.945) under stratified five-fold cross-validation. This result demonstrated strong discriminative capability for distinguishing between patients with and without heart disease presence within the dataset.

The random forest benchmark model was additionally implemented to evaluate the effect of nonlinear ensemble learning on predictive performance. Although ensemble-based

approaches can capture nonlinear interactions among cardiovascular variables, the interpretable logistic regression model achieved comparable and slightly superior predictive performance in the present study. These findings support the use of transparent machine learning models in healthcare applications where explainability and clinical interpretability are important.

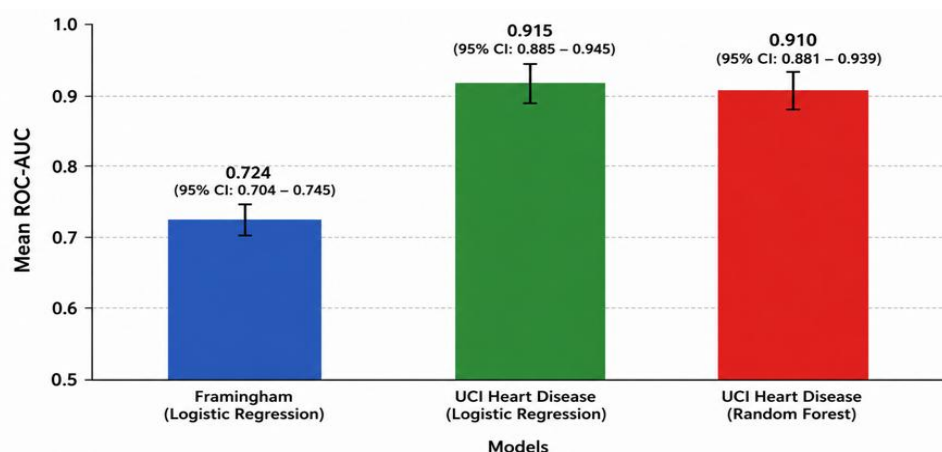
The results therefore reinforced one of the central motivations of the study: achieving an appropriate balance between predictive capability and clinical interpretability. While more complex models may improve numerical performance in some cases, transparent models may offer greater practical value in healthcare environments where explainability is essential. The ROC curves and discriminative performance comparison of the evaluated models are presented in Fig. 7.

Fig. 7. ROC curves and AUC comparison for all models



Mean ROC-AUC performance comparison across the evaluated cardiovascular prediction models is illustrated in Fig. 8.

Fig. 8. Mean ROC-AUC performance comparison across evaluated cardiovascular prediction models



H. Comparative Analysis of the Prediction Models

A comparison between the two modeling settings revealed important differences in both predictive performance and clinical interpretation. The Framingham-based model utilized a lightweight feature set and focused on long-term cardiovascular risk estimation, whereas the UCI-based model incorporated a more extensive collection of cardiovascular indicators for direct heart disease classification.

The higher ROC-AUC achieved by the UCI-based model can likely be attributed to the inclusion of clinically detailed variables such as chest pain type, electrocardiographic results, exercise-induced angina, and ST-segment measurements. These features provide stronger immediate indicators of cardiovascular abnormalities compared to the smaller preventive-risk-oriented feature set used in the Framingham model.

However, the Framingham-based framework demonstrated important strengths in simplicity, accessibility, and interpretability. Because the model relied only on basic clinical measurements and lifestyle-related factors, it may be more suitable for early screening or preventive awareness settings where advanced diagnostic variables are unavailable.

The comparative analysis therefore highlighted that predictive performance alone should not be the sole criterion for evaluating healthcare machine learning systems. Transparency, explainability, and clinical usability remain equally important considerations in biomedical informatics applications.

I. Feature Interpretation and Clinical Relevance

An important component of the proposed framework involved interpretation of model behavior using clinically meaningful feature analysis. Logistic regression coefficients and corresponding odds ratios were examined to identify variables associated with increased cardiovascular risk.

Odds-ratio analysis demonstrated that age (OR=1.723, 95% CI: 1.561-1.901, $p<0.0001$) was the strongest predictor of cardiovascular risk. Elevated systolic blood pressure (OR=1.466, 95% CI: 1.341-1.604, $p<0.0001$), male sex (OR=1.324, 95% CI: 1.209-1.450, $p<0.0001$), current smoking (OR=1.212, 95% CI: 1.103-1.332, $p=0.0001$), and diabetes (OR=1.129, 95% CI: 1.056-1.208, $p=0.0004$) were also statistically significant predictors. Body mass index was not statistically significant (OR=1.028, 95% CI: 0.939-1.125, $p=0.5532$).

Table V

Clinically Interpretable Risk Factors and Their Contribution to Cardiovascular Prediction

Clinical Variable	Odds Ratio	95% Confidence Interval	P-value	Clinical Interpretation
Male Sex	1.324	1.209-1.450	<0.0001	Increased cardiovascular risk
Age	1.723	1.561-1.901	<0.0001	Strongest predictor

Systolic Blood Pressure	1.466	1.341-1.604	<0.0001	Significant risk factor
Current Smoking	1.212	1.103-1.332	0.0001	Significant risk factor
Diabetes	1.129	1.056-1.208	0.0004	Significant risk factor
BMI	1.028	0.939-1.125	0.5532	Not statistically significant

Similarly, the UCI-based model identified clinically recognized cardiovascular indicators including exercise-induced angina, elevated cholesterol levels, abnormal chest pain characteristics, and ST-segment depression as important contributors to heart disease prediction. The ability of the model to align with known clinical relationships improved confidence in the interpretability of generated outputs.

Because logistic regression preserves direct mathematical relationships between variables and prediction probabilities, the resulting explanations could be communicated in a medically understandable manner without requiring highly complex post-hoc interpretation methods.

J. Human-Readable Explainability Outputs

Beyond numerical prediction scores, the framework generated human-readable explanations intended to improve accessibility and transparency of risk communication. After probability estimation, the system categorized outputs into low-, moderate-, or high-risk groups and provided explanations describing major contributing factors and protective indicators.

For example, elevated blood pressure, obesity, smoking status, diabetes, and abnormal cardiovascular indicators were explicitly identified as factors increasing predicted risk. At the same time, favorable physiological conditions such as healthy blood pressure ranges, absence of smoking, and better exercise tolerance were communicated as protective factors.

The framework additionally provided general lifestyle-oriented recommendations related to physical activity, dietary habits, blood pressure management, smoking cessation, and routine medical evaluation. These explanation components were designed to improve interpretability while avoiding direct diagnostic claims.

The experimental results demonstrated that explainability-oriented communication can be integrated into cardiovascular prediction systems without substantially increasing methodological complexity.

K. Summary of Experimental Findings

Overall, the experimental analysis demonstrated that interpretable machine learning models can provide meaningful cardiovascular prediction performance while maintaining transparency and clinically understandable outputs. The Framingham-based model achieved moderate but clinically relevant predictive capability using a lightweight and accessible feature set, whereas the UCI-based model achieved stronger discriminative performance using a more detailed collection of cardiovascular indicators.

The study findings further suggest that explainability and ethical communication can be integrated directly into machine learning-based healthcare frameworks rather than treated as

secondary considerations. By combining transparent predictive modeling with human-readable explanation strategies, the proposed framework contributes toward more interpretable and human-centered cardiovascular risk assessment systems.

VI. Discussion

The present study demonstrates that clinically interpretable machine learning models can provide meaningful cardiovascular risk prediction while maintaining transparency and accessibility. The Framingham-based model achieved moderate predictive performance using a lightweight set of routinely available clinical variables, whereas the UCI-based model demonstrated stronger discriminative capability through inclusion of more detailed cardiovascular indicators. These findings are consistent with previous work highlighting the growing role of machine learning in healthcare (Beam & Kohane, 2018). Furthermore, the results suggest that clinically accessible variables may still support useful risk estimation, particularly in preventive and educational healthcare settings.

An important objective of this work was to emphasize explainability alongside predictive capability. Many modern machine learning approaches achieve strong numerical performance but provide limited insight into how predictions are generated. In contrast, the proposed framework utilized logistic regression because its coefficients and odds ratios can be interpreted directly in relation to recognized cardiovascular risk factors. This transparency improves the ability to communicate predictions in clinically understandable terms and may support greater trust among clinicians and patients.

The study also highlights the importance of ethical communication in healthcare artificial intelligence. Rather than presenting outputs as definitive diagnostic conclusions, the proposed system framed predictions as probabilistic risk estimates intended for supportive and educational use. Human-readable explanations and medical disclaimers were incorporated to encourage responsible interpretation of model outputs. Such transparency is important because healthcare AI systems may influence patient understanding and clinical decision-making.

A. Limitations of the Study

Despite the encouraging findings obtained in the present study, several limitations should be acknowledged. First, the proposed framework was developed and evaluated using publicly available cardiovascular datasets that represent specific populations and clinical settings. Consequently, the generalizability of the developed models to broader, geographically diverse, or contemporary patient populations remains uncertain.

Second, the study relied primarily on structured clinical variables and did not incorporate additional sources of information such as longitudinal electronic health records, medical imaging data, laboratory biomarkers, wearable sensor measurements, or genomic information. Inclusion of multimodal healthcare data may further improve predictive capability and clinical applicability.

Third, although external validation was performed using an independent cardiovascular dataset comprising approximately 70,000 participants, additional validation using geographically diverse populations, prospective clinical cohorts, and real-world healthcare

settings remains necessary to further establish the robustness and generalizability of the proposed framework.

Fourth, the explainability strategy adopted in this work relied primarily on the inherent interpretability of logistic regression models, clinically guided explanation rules, and SHAP-based explainability analysis. While these approaches provide transparent and human-readable explanations of cardiovascular risk prediction, additional explainable artificial intelligence techniques such as LIME and counterfactual explanations may provide complementary insights into model behavior and support more comprehensive interpretability assessment.

Finally, although calibration analysis, fairness assessment, and external validation were performed using calibration curves, Brier score evaluation, sex-based performance comparison, and independent dataset validation, additional investigations involving uncertainty quantification, clinician-centered usability evaluation, and prospective clinical studies are required before practical clinical deployment can be considered. While the present study demonstrated favorable fairness characteristics across sex categories, future investigations should evaluate additional demographic groups and broader population cohorts.

Despite these limitations, the present study demonstrates that lightweight, interpretable, and ethically framed machine learning approaches can provide meaningful cardiovascular risk assessment while preserving transparency and clinical usability.

B. Future Research Directions:

Another important consideration involves probabilistic calibration of cardiovascular risk estimates. Although ROC-AUC analysis demonstrated meaningful discriminative capability, additional calibration-oriented evaluation may further improve the clinical reliability of predicted probability scores.

Future work may involve external validation using larger and more diverse cardiovascular datasets, integration of advanced explainable AI methods, and incorporation of longitudinal clinical information, imaging data, and wearable sensor measurements. Further investigation of fairness, calibration, and clinically deployable decision-support interfaces may also strengthen the practical applicability of interpretable cardiovascular prediction systems. Future research should also investigate demographic fairness and population-level generalizability of interpretable cardiovascular prediction systems. Since publicly available datasets may contain demographic imbalance and population-specific clinical characteristics, fairness-aware evaluation across age groups, sex categories, and diverse healthcare settings remains an important direction for trustworthy healthcare AI research.

The lightweight nature of the proposed framework may additionally support implementation in resource-constrained healthcare environments where computational infrastructure and advanced diagnostic systems are limited. Because the framework primarily relies on clinically accessible variables and computationally efficient models, it may provide advantages for preventive healthcare screening and educational decision-support applications.

Overall, the findings suggest that transparent and ethically framed machine learning models can support meaningful cardiovascular risk assessment while preserving clinical interpretability and responsible communication principles.

VII. Conclusion

This study presented an explainable and clinically interpretable machine learning framework for cardiovascular risk prediction using the Framingham Heart Study dataset and the UCI Heart Disease dataset. The proposed framework was designed not only to generate predictive estimates but also to emphasize transparency, ethical communication, and clinically meaningful interpretation of cardiovascular risk factors.

Two distinct cardiovascular prediction settings were investigated within the study. The Framingham-based model focused on long-term cardiovascular risk estimation using a lightweight set of clinically accessible variables, while the UCI-based model addressed heart disease probability assessment using a broader range of cardiovascular indicators. Logistic regression was selected as the primary predictive approach because of its interpretability, computational simplicity, and ability to provide clinically understandable coefficient and odds-ratio analysis. A random forest model was additionally evaluated as a nonlinear benchmark for comparative analysis.

Experimental evaluation demonstrated that interpretable machine learning models can achieve meaningful predictive performance while preserving transparency of decision-making processes. The Framingham-based model achieved moderate but clinically relevant predictive capability, whereas the UCI-based model demonstrated stronger discriminative performance due to the inclusion of more detailed cardiovascular variables. Beyond predictive performance, the framework generated human-readable explanations describing contributing risk factors, protective indicators, and practical lifestyle-oriented recommendations.

An important contribution of the study lies in its emphasis on ethical and responsible healthcare AI communication. Rather than presenting predictions as definitive clinical diagnoses, the framework treated outputs as probabilistic risk estimates intended to support awareness and research-oriented decision assistance. Medical disclaimers and transparent explanation strategies were incorporated to encourage appropriate interpretation of model outputs.

The findings of this work suggest that lightweight and interpretable machine learning systems may provide valuable support for cardiovascular risk assessment, particularly in settings where transparency, accessibility, and clinician trust are important considerations. At the same time, the study acknowledges important limitations related to dataset generalizability, external clinical validation, and the scope of explainability methods used.

Future research may further strengthen the framework through the integration of advanced explainable AI techniques, larger and more diverse datasets, multimodal healthcare information, and prospective clinical evaluation. Overall, the present study highlights the potential of transparent and human-centered machine learning approaches for supporting trustworthy and ethically responsible cardiovascular healthcare technologies. To promote transparency, reproducibility, and further research, all implementation code used in this study has been made publicly available through a GitHub repository.

Declarations

Ethics approval and consent to participate

This study used only publicly available, anonymized datasets (Framingham Heart Study dataset, UCI Heart Disease dataset, and Kaggle Cardiovascular Disease dataset). No human participants were recruited, no patient intervention was performed, and no identifiable personal data were collected by the authors. Therefore, ethics committee approval and informed consent were not required.

Consent for publication

Not applicable.

Clinical trial registration

This study did not involve a clinical trial; therefore, trial registration was not applicable.

Availability of data and materials

The datasets analyzed during the current study are publicly available from their original sources:

- Framingham Heart Study dataset
- UCI Heart Disease dataset (UCI Machine Learning Repository)
- Kaggle Cardiovascular Disease dataset

The source code used for data preprocessing, model development, performance evaluation, explainability analysis, calibration analysis, fairness assessment, and external validation is publicly available at:

<https://github.com/kittudulam/Explainable-Cardiovascular-Risk-Prediction>

The datasets are not redistributed with the source code because they are subject to their respective licensing and distribution terms.

Competing interests

The author declares that there are no competing interests.

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Authors' contributions

Sai Keerthana Dulam conceived the study, designed the methodology, implemented the machine learning models, performed data analysis, interpreted the results, and wrote the manuscript. The author read and approved the final manuscript.

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